

DDTA - Scheda di lettura compilata

SRC-0048 - ISO/IEC/IEEE 42010:2011 full-text regression

REVIEW-ONLY: historical full-text source; 2022 edition remains separate ACCESS-LIMITED SRC-0043.

Identita bibliografica e accesso

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1. Research problem and contribution

The standard governs architecture descriptions, viewpoints, frameworks and ADLs. It sharply distinguishes the abstract architecture of a system from the work product used to express it. It does not define what a system is and does not prescribe one method, notation or medium.

2. Starting artifacts and representation

The conceptual model separates system-of-interest, architecture, architecture description, stakeholder, concern, viewpoint, view, architecture model, model kind, correspondence, decision and rationale. Annex B states that a model-kind metamodel can define entities, attributes, relationships and constraints. This supports the BA0-R extraction lens but does not determine the DDTA BAE classes.

3. Traceability, change, automation and human role

Correspondences explicitly relate architecture-description elements and can be governed by rules. They can express consistency, traceability, composition, refinement, dependency and model transformation, including links to requirements. Key architecture decisions can carry identity, affected elements, concerns, owner, rationale, constraints, alternatives, consequences, timestamps and source citations. Architecture descriptions can aggregate several life-cycle work products and serve as input to simulation, generation and analysis.

4. Main DDTA challenge

Earlier projection claim is too strong as prior-art claim

Annex A describes both a synthetic approach, where separately constructed views are integrated through correspondences, and a projective approach, where views are extracted from an underlying repository. ISO 42010 supports either. DDTA may still choose a canonical projective representation, but BA0/BA4 must justify it as a DDTA design decision rather than treat it as mandatory architecture semantics.

Hypothesis effects

Architecture != description	STRONGLY SUPPORTED.
View != viewpoint	STRONGLY SUPPORTED.
One canonical repository mandatory	REJECTED as an ISO prior-art claim.
Projective DDTA views	CANDIDATE design choice, not external invariant.
Partial diagram == architecture view	REJECTED if ISO terminology is adopted; partial content fits architecture model/projection better.
Correspondence semantics	STRONGLY EVIDENCED prior art.
Decision/rationale	STRONGLY EVIDENCED prior art.
ISO 42010 supplies BAE ontology	REJECTED; it explicitly does not define system nature.

Citation-ready evidence

PDF	Ruolo	Sezione	DDTA interpretation
8	foundation	Terms	Architecture, AD, view, viewpoint and model kind are distinct.
10	challenge	4.2.1	System nature is intentionally not defined; environment matters.
11	foundation	4.2.2	Architecture is abstract; AD is a work product; method/notation/media are open.
12	foundation	4.2.4–4.2.5	Viewpoint governs view; models are governed by model kinds.
13	comparison	4.2.6–4.2.7	Correspondence and decision/rationale are first-class architecture-description concepts.
20–21	comparison	5.6–5.7	Models can be shared across views; correspondences support consistency and traceability.
22	comparison	5.8	Key decisions may record affected elements, alternatives, consequences, timestamps and citations.
24	foundation	Clause 7	Viewpoint conventions can include analysis methods and consistency operations.
28	challenge	A.4	Synthetic and projective view construction are both supported.
34	foundation	B.2.6	Model-kind metamodel lens: entities, attributes, relationships, constraints.
40	comparison	C.4	RM-ODP places information, interfaces, interactions and channel structure in different viewpoints.

Chapter regression and status

No Chapter 3 or Chapter 4 reopen trigger is found. The important downstream correction belongs to BA0/BA4: canonical-projective view generation is a candidate DDTA choice that must be justified, and ‘view’ terminology should distinguish holistic concern-oriented views from partial models/renderings.

Governance note

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